Named Entity Recognition

Case Study: Information Extraction from News Text

Natural Language Understanding & Generation | April 2026

Objective

Named Entity Recognition (NER) is the task of locating and classifying named entities in unstructured text into predefined categories such as person, organisation, and location. This report compares three implementation approaches of increasing complexity - a statistical model, a pre-trained transformer, and a fine-tuned transformer - evaluated on the CoNLL-2003 news benchmark.

Dataset

CoNLL-2003 is the standard NER benchmark, built from Reuters newswire text annotated with four entity types in IOB2 format: PER (person), ORG (organisation), LOC (location), and MISC (miscellaneous).

Split	Sentences	Tokens
Train	14,041	203,621
Validation	3,250	51,362
Test	3,684	46,435

Methodology

Part 1 - spaCy

The en_core_web_sm pipeline uses a CNN-based model trained on OntoNotes 5, which covers 18 entity types. For this study, the relevant types - PERSON, ORG, GPE/LOC, and DATE - map closely to the four CoNLL-2003 labels. The model runs entirely offline with no GPU requirement.

Part 2 - Pre-trained BERT

The dbmdz/bert-large-cased-finetuned-conll03-english checkpoint is loaded directly via the HuggingFace pipeline with aggregation_strategy='simple' to merge sub-word token predictions into complete entity spans.

Part 3 - Fine-tuning BERT-Base

bert-base-cased is fine-tuned using the HuggingFace Trainer API with the following configuration: learning rate 2e-5 with 10% warmup, AdamW optimiser, weight decay 0.01, batch size 16, and 3 epochs. Because BERT's WordPiece tokeniser splits words into sub-word tokens, labels must be re-aligned: the first sub-token of each word receives the NER label and continuation tokens are assigned -100, a HuggingFace convention that excludes them from the loss computation. Evaluation uses strict entity-level F1 via seqeval.

Results

Approach	Model	Size	F1 (test)
spaCy	en_core_web_sm	12 MB	~85

BERT (pre-trained)	bert-large-cased-finetuned	1.3 GB	~92
BERT (fine-tuned)	bert-base-cased	420 MB	~90

F1 scores are entity-level (strict matching) on the CoNLL-2003 test set.

Sample Output

Entities extracted by the pre-trained BERT pipeline on the four sample sentences. Note: 'Washington D.C.' was split into two tokens by the model's WordPiece tokeniser and is excluded from this table to avoid misrepresentation.

Entity	Label	Confidence
Tesla	ORG	0.9969
Austin	LOC	0.9990
Narendra Modi	PER	0.9952
Hiroshima	LOC	0.9989
Andy Jassy	PER	0.9957
Amazon	ORG	0.9994
International Monetary Fund	ORG	0.9988

Key Findings

- spaCy is over 100x smaller than BERT-Large and runs offline, making it suitable for low-latency or resource-constrained deployments at the cost of around 7 F1 points.

- The pre-trained BERT-Large model achieves the highest accuracy with no additional training, demonstrating strong transfer from large-scale pre-training to NER.

- Fine-tuned BERT-Base reaches ~90 F1 and offers a practical balance between model size, inference speed, and accuracy for production use.

- Warmup scheduling and weight decay are important for stable transformer fine-tuning and prevent large gradient updates in early training steps.

- Sub-word tokenisation requires label alignment: only the first sub-token of each word receives the NER label; continuation tokens are masked from the loss.

Deployment

Inference is wrapped in a NERModel class that applies a confidence threshold (min_score=0.80) to filter low-confidence predictions and supports batch processing via the HuggingFace pipeline's native batching. Example output:

```
Reliance Industries   ORG    0.9996
Google                   ORG    0.9994
Gemini AI                 ORG    0.9912
Geneva                     LOC    0.9998
```

Conclusion

All three approaches successfully extract named entities from news text. spaCy is the best choice when speed and offline operation are priorities. Pre-trained BERT-Large gives the highest accuracy out of the box. Fine-tuned BERT-Base is the recommended option for production systems where model size and inference cost need to be controlled. The HuggingFace ecosystem provides a unified API across all approaches, reducing the effort required to switch between models.